

ETHICAL HACKING TASK 01

Information Gathering & Reconnaissance Report

Target: VIT University

URL: <https://www.vit.ac.in/>

Submitted by: Lakshman Govindan

Date: June 2026

White Band Associates Internship

Part A: Target Information

Website Name: VIT University (Vellore Institute of Technology)

URL: <https://www.vit.ac.in/>

Type: Deemed University — Educational Institution

Location: Vellore, Tamil Nadu, India

Reason for Choosing:

VIT University is a well-known public Indian deemed university with a large online presence. It is an appropriate and ethical target for passive reconnaissance as it is a publicly accessible educational institution. No systems were accessed or exploited — this exercise is purely educational.

Part B: WHOIS Lookup

Performed using whois.com. The .ac.in domain is a restricted academic TLD in India managed by ERNET India.

Field	Value
Domain Name	vit.ac.in
Registered On	2003-06-30
Expires On	2028-06-30
Last Updated	2019-05-18
Status	ok
Registrar	ERNET India (IANA ID: 800068)
Registrar URL	http://www.ernet.in
Abuse Email	tejalt@eis.ernet.in
Abuse Phone	+91.1123358248
Registrant State	Tamil Nadu
Registrant Country	IN
Name Servers	ns-1772.awsdns-29.co.uk, ns-389.awsdns-48.com, ns-865.awsdns-44.net, ns-1067.awsdns-05.org
DNSSEC	Unsigned
Registrant Contact	REDACTED FOR PRIVACY (ICANN compliance)

Screenshot

The screenshot shows the whois.com website with the domain vit.ac.in selected. The page layout includes a navigation bar at the top with links for Domains, Hosting, Websites, Email, Security, Whois, and Deals. Below the navigation bar, the domain vit.ac.in is displayed with a 'Updated 5 days ago' status. The main content area is divided into three sections: Domain Information, Registrar Information, and Registrant Contact. The Domain Information section lists the domain name, registration date (2003-06-30), expiration date (2028-06-30), last update date (2019-05-18), status (ok), and name servers (ns-1772.awsdns-29.co.uk, ns-389.awsdns-48.com, ns-865.awsdns-44.net, ns-1067.awsdns-05.org). The Registrar Information section lists the registrar (ERNET India), IANA ID (800068), URL (http://www.ernet.in), abuse email (tejalt@eis.ernet.in), and abuse phone (+91.1123358248). The Registrant Contact section lists the state (Tamil Nadu) and country (IN). To the right of the domain information, there is a section titled 'Interested in similar domains?' with a list of domains and 'Buy Now' buttons. Below this, there are two promotional banners: one for .space domains with a price of \$1.18 and another for .biz domains with a price of \$8.88.

Domain Information	Updated 5 days ago	Interested in similar domains?
Domain: vit.ac.in		vitllc.com Buy Now
Registered On: 2003-06-30		gyllt.com Buy Now
Expires On: 2028-06-30		vittechnologies.com Buy Now
Updated On: 2019-05-18		vitlaw.com Buy Now
Status: ok		vitcloud.net Buy Now
Name Servers: ns-1772.awsdns-29.co.uk ns-389.awsdns-48.com ns-865.awsdns-44.net ns-1067.awsdns-05.org		vesit.net Buy Now
Registrar Information		.space Sale \$29.88 \$1.18 BUY NOW *while stocks last
Registrar: ERNET India		On Sale! .biz \$12.99 \$8.88 *while stocks last
IANA ID: 800068		
URL: http://www.ernet.in		
Abuse Email: tejalt@eis.ernet.in		
Abuse Phone: +91.1123358248		
Registrant Contact		
State: Tamil Nadu		
Country: IN		
Raw Whois Data		

Screenshot 1: whois.com — WHOIS lookup for vit.ac.in showing registration details

Part C: DNS Enumeration

Performed using dnschecker.org.

A Records

IP Address	Notes
122.184.65.22	Tata Communications — Indian ISP hosting

MX Records

Priority	Mail Server	Notes
1	aspmx.l.google.com	Primary — Google Workspace
5	alt1.aspmx.l.google.com	Google Workspace alternate
5	alt2.aspmx.l.google.com	Google Workspace alternate
10	alt3.aspmx.l.google.com	Google Workspace backup
10	alt4.aspmx.l.google.com	Google Workspace backup

NS Records

Name Server	Provider
ns-389.awsdns-48.com	Amazon Route 53
ns-865.awsdns-44.net	Amazon Route 53
ns-1067.awsdns-05.org	Amazon Route 53
ns-1772.awsdns-29.co.uk	Amazon Route 53

TXT Records

TXT Record	Purpose
v=spf1 a:smtp-relay.gmail.com a:ciscoesa.vit.ac.in include:_spf.google.com include:spf.protection.outlook.com include:amazonses.com -all	SPF — authorizes Google, Microsoft, Amazon SES, Cisco ESA to send email
_globalsign-domain-verification=...	GlobalSign SSL certificate verification
google-site-verification=GhsNRGhW7NS7TwDPpOH6tv6C...	Google Search Console ownership verification
google-site-verification=47M3SdmhHUx73lZvVbawAtJMr...	Secondary Google Search Console verification
_escow7ziteo3a23k8zuzqbugzxcg5mkw	Escow domain verification token

Screenshots

vit.ac.in

MX

Search

CD Flag

Refresh: 20 sec.

San Francisco CA, United States OpenDNS	1:aspmx.l.google.com. 5:alt1.aspmx.l.google.com. 5:alt2.aspmx.l.google.com. 10:alt3.aspmx.l.google.com. 10:alt4.aspmx.l.google.com.	✓
Mountain View CA, United States Google	10:alt4.aspmx.l.google.com. 10:alt3.aspmx.l.google.com. 1:aspmx.l.google.com. 5:alt1.aspmx.l.google.com. 5:alt2.aspmx.l.google.com.	✓
Berkeley, US Quad9	5:alt1.aspmx.l.google.com. 10:alt3.aspmx.l.google.com. 1:aspmx.l.google.com. 10:alt4.aspmx.l.google.com. 5:alt2.aspmx.l.google.com.	✓
United States CenturyLink	10:alt3.aspmx.l.google.com. 5:alt1.aspmx.l.google.com. 5:alt2.aspmx.l.google.com. 1:aspmx.l.google.com. 10:alt4.aspmx.l.google.com.	✓
Alpharetta, United States ZCORUM	5:alt2.aspmx.l.google.com. 10:alt3.aspmx.l.google.com. 10:alt4.aspmx.l.google.com. 5:alt1.aspmx.l.google.com. 1:aspmx.l.google.com.	✓
Ashburn, United States NeuStar	10:alt4.aspmx.l.google.com. 5:alt1.aspmx.l.google.com. 5:alt2.aspmx.l.google.com. 1:aspmx.l.google.com. 10:alt3.aspmx.l.google.com.	✓

Whether you have recently changed your DNS records are propagated globally is essential. DNS Name System records against a selected list of DNS lookup for any hostname or domain, and check DNS records are fully propagated.

DNS Propagation Map b

Screenshot 2: DNS A Record — 122.184.65.22 (Tata Communications)

vit.ac.in

NS

Search

CD Flag

Refresh: 20 sec.

San Francisco CA, United States OpenDNS	ns-389.awsdns-48.com. ns-865.awsdns-44.net. ns-1067.awsdns-05.org. ns-1772.awsdns-29.co.uk.	✓
Mountain View CA, United States Google	ns-1067.awsdns-05.org. ns-865.awsdns-44.net. ns-1772.awsdns-29.co.uk. ns-389.awsdns-48.com.	✓
Berkeley, US Quad9	ns-865.awsdns-44.net. ns-389.awsdns-48.com. ns-1067.awsdns-05.org. ns-1772.awsdns-29.co.uk.	✓
United States CenturyLink	ns-1772.awsdns-29.co.uk. ns-865.awsdns-44.net. ns-389.awsdns-48.com. ns-1067.awsdns-05.org.	✓
Alpharetta, United States ZCORUM	ns-1067.awsdns-05.org. ns-1772.awsdns-29.co.uk. ns-865.awsdns-44.net. ns-389.awsdns-48.com.	✓
Ashburn, United States NeuStar	ns-865.awsdns-44.net. ns-1067.awsdns-05.org. ns-1772.awsdns-29.co.uk. ns-389.awsdns-48.com.	✓
San Francisco, US Quad9	ns-865.awsdns-44.net. ns-389.awsdns-48.com. ns-1067.awsdns-05.org. ns-1772.awsdns-29.co.uk.	✓
Burnaby, Canada Fortinet Inc	ns-1067.awsdns-05.org. ns-389.awsdns-48.com. ns-865.awsdns-44.net.	✓

Whether you have recently changed your records are propagated globally is essential. DNS Name System records against a selected lookup for any hostname or domain, and records are fully propagated.

DNS Propagation Map

Screenshot 3: DNS MX Records — All 5 records point to Google Workspace

vit.ac.in TXT Search

CD Flag Refresh: 20 sec.

San Francisco CA, United States
OpenDNS ⓘ

```

- all\
_4z2ycmu8ctx1vi6vo3pm4dqaj2motx3
_aefuvbli7hxs8yz5ehwexgrb5qxaw
_escow7ziteo3a23k8zuzqbugxg5mkw
google-site-
verification=47M3SdmhHUX73IZvVbawAtJMrff6DPwG
TputfmSkaNs
google-site-
verification=GhsNRGhW7NS7TwDPpOH6tv6C_Ql3vSW
V42MbGgzbOeg
_globalsign-domain-
verification=25CgQLQdRBzs3ZN_RRWodGrcmKwSO5VB
te8k7XF5Z0W
v=spf1 a:smtp-relay.gmail.com a:ciscoesa.vit.ac.in
a:ciscoesa2.vit.ac.in include:_spf.google.com
include:_spf.protection.outlook.com
include:_spf.vit.ac.in include:_spf2.vit.ac.in
include:amazonses.com -all
_escow7ziteo3a23k8zuzqbugxg5mkw
_globalsign-domain-
verification=25CgQLQdRBzs3ZN_RRWodGrcmKwSO5VB
te8k7XF5Z0W
google-site-
verification=GhsNRGhW7NS7TwDPpOH6tv6C_Ql3vSW
V42MbGgzbOeg
google-site-
verification=47M3SdmhHUX73IZvVbawAtJMrff6DPwG
TputfmSkaNs
_aefuvbli7hxs8yz5ehwexgrb5qxaw
_4z2ycmu8ctx1vi6vo3pm4dqaj2motx3
v=spf1 a:smtp-relay.gmail.com a:ciscoesa.vit.ac.in
a:ciscoesa2.vit.ac.in include:_spf.google.com
include:_spf.protection.outlook.com
include:_spf.vit.ac.in include:_spf2.vit.ac.in

```

Mountain View CA, United States
Google ⓘ

Screenshot 4: DNS NS Records — All 4 Amazon Route 53 name servers

vit.ac.in TXT Search

CD Flag Refresh: 20 sec.

San Francisco CA, United States
OpenDNS ⓘ

```

- all\
_4z2ycmu8ctx1vi6vo3pm4dqaj2motx3
_aefuvbli7hxs8yz5ehwexgrb5qxaw
_escow7ziteo3a23k8zuzqbugxg5mkw
google-site-
verification=47M3SdmhHUX73IZvVbawAtJMrff6DPwG
TputfmSkaNs
google-site-
verification=GhsNRGhW7NS7TwDPpOH6tv6C_Ql3vSW
V42MbGgzbOeg
_globalsign-domain-
verification=25CgQLQdRBzs3ZN_RRWodGrcmKwSO5VB
te8k7XF5Z0W
v=spf1 a:smtp-relay.gmail.com a:ciscoesa.vit.ac.in
a:ciscoesa2.vit.ac.in include:_spf.google.com
include:_spf.protection.outlook.com
include:_spf.vit.ac.in include:_spf2.vit.ac.in
include:amazonses.com -all
_escow7ziteo3a23k8zuzqbugxg5mkw
_globalsign-domain-
verification=25CgQLQdRBzs3ZN_RRWodGrcmKwSO5VB
te8k7XF5Z0W
google-site-
verification=GhsNRGhW7NS7TwDPpOH6tv6C_Ql3vSW
V42MbGgzbOeg
google-site-
verification=47M3SdmhHUX73IZvVbawAtJMrff6DPwG
TputfmSkaNs
_aefuvbli7hxs8yz5ehwexgrb5qxaw
_4z2ycmu8ctx1vi6vo3pm4dqaj2motx3
v=spf1 a:smtp-relay.gmail.com a:ciscoesa.vit.ac.in
a:ciscoesa2.vit.ac.in include:_spf.google.com
include:_spf.protection.outlook.com
include:_spf.vit.ac.in include:_spf2.vit.ac.in

```

Mountain View CA, United States
Google ⓘ

Screenshot 5: DNS TXT Records — SPF, GlobalSign, Google verification tokens

Part D: Website Technology Identification

Technology identification was performed through DNS analysis, robots.txt examination, and URL pattern analysis.

Category	Technology	Evidence
Web Hosting	Tata Communications	IP 122.184.65.22 belongs to Tata Communications (Indian ISP)
DNS Management	Amazon Route 53	All four name servers are AWS Route 53 (awsdns-*)
CMS	WordPress	/wp-admin/ and /xmlrpc.php found in robots.txt
Email (Primary)	Google Workspace	All 5 MX records point to Google (aspmx.l.google.com)
Email (Backup)	Microsoft 365	SPF includes spf.protection.outlook.com
Email (Transactional)	Amazon SES	SPF includes amazonses.com
Email (Security Gateway)	Cisco ESA	SPF includes ciscoesa.vit.ac.in
SSL Certificate	GlobalSign	GlobalSign domain verification token in TXT records
Search Verification	Google Search Console	Multiple google-site-verification TXT tokens
CDN	None detected	Single IP, no Cloudflare or Akamai headers
LMS Subdomain	Moodle (vitolcc.vit.ac.in)	VIT Online Learning Centre runs Moodle LMS

Technology Summary

VIT University's website is built on WordPress CMS, hosted on Tata Communications infrastructure in India with DNS managed by Amazon Route 53. Email infrastructure is notably complex — using Google Workspace as the primary email provider, with Microsoft 365, Amazon SES, and Cisco ESA as additional providers for backup, transactional, and security gateway purposes respectively. GlobalSign SSL certificate provides HTTPS. No CDN was detected.

Part E: HTTP Security Headers

HTTP security headers were checked via securityheaders.com. HTTPS is active on the site.

Security Header	Present	Purpose
Content-Security-Policy (CSP)	Not Detected	Prevents XSS attacks by specifying which content sources are trusted
X-Frame-Options	Not Detected	Prevents the page from being embedded in iframes (clickjacking protection)
X-Content-Type-Options	Not Detected	Prevents MIME type sniffing by the browser
Strict-Transport-Security (HSTS)	Not Detected	Forces browsers to always use HTTPS even if user types http://
Referrer-Policy	Not Detected	Controls how much referrer info is included with requests
Permissions-Policy	Not Detected	Controls access to browser features (camera, mic, geolocation)

Observations

- HTTPS is active — GlobalSign SSL certificate is in use.
- None of the major HTTP security headers were detected — common in WordPress-based university websites.
- Absence of CSP and X-Frame-Options leaves potential for XSS and clickjacking attacks.
- No HSTS means users could potentially be downgraded to HTTP connections.

Part F: Robots.txt & Sitemap Analysis

robots.txt

URL: <https://www.vit.ac.in/robots.txt> — PRESENT

Rule	Finding
SemrushBot, MJ12bot, GermCrawler, Sogou	Disallow: / — these SEO/spam bots blocked entirely
Googlebot, Googlebot-News etc.	Allow: *.js and *.css only
Disallow: /xmlrpc.php	Reveals WordPress — xmlrpc.php is a known attack vector for brute force
Disallow: /admin/	Admin panel path exposed
Disallow: /wp-admin/	WordPress admin path — confirms WordPress CMS
Disallow: /comment/reply/	Comment reply path blocked
Disallow: /search/	Search page blocked
Disallow: /cgi-bin/	Server-side scripts directory blocked
Disallow: /school/allfaculty/	Faculty listing page blocked
Disallow: /noteable-alumni/	Notable alumni page blocked
Allow: /admissions/overview	Admissions overview allowed for indexing
Allow: /all-courses/ug and /all-courses/pg	Course listing pages allowed

Key Finding:

robots.txt reveals /wp-admin/ and /xmlrpc.php — confirming WordPress CMS. The xmlrpc.php file is a well-known WordPress attack vector that can be exploited for brute force and DDoS amplification attacks. While blocking it in robots.txt hides it from search engines, the file itself may still be accessible.

sitemap.xml

URL: <https://www.vit.ac.in/sitemap.xml> — PRESENT

Item	Finding
File Present	Yes
Generator	Free Online Sitemap Generator (www.xml-sitemaps.com)
Homepage	https://vit.ac.in/ (priority: 1.0)
Last Modified	2024-11-05
Indexed pages include	/events-photos-and-videos, /admissions/international/overview/, /about-vit, /about/vision-mission, /vit-milestones, /about/leadership, /governance, /about/administrative-offices, /about/infrastructure, /about/sustainability, /true-green and more

Screenshots

```
User-agent: SearushBot
Disallow: /

User-agent: MJ12bot
Disallow: /

User-agent: GerniCrawler
Disallow: /

User-agent: Sogou
Disallow: /

User-Agent: Googlebot
User-Agent: Googlebot-News
User-Agent: Googlebot-Image
User-Agent: Googlebot-Video
User-Agent: Googlebot-Mobile
User-Agent: Mediapartners-Google
Allow: *.js
Allow: *.css

User-agent: *
Disallow: /*?utm_*
Disallow: /xmlrpc.php
Disallow: /admin/
Disallow: /wp-admin/
Disallow: /comment/reply/
Disallow: /search/
Disallow: /cgi-bin/
Allow: /admissions/overview
Allow: /admissions/programmes-offered
Allow: /all-courses/ug
Allow: /all-courses/pg
Disallow: /school/allfaculty/
Disallow: /notable-alumni/
Disallow: /category/
```

Screenshot 6: robots.txt — reveals /wp-admin/ and /xmlrpc.php confirming WordPress CMS

```
This XML file does not appear to have any style information associated with it. The document tree is shown below.
<?xml version="1.0" encoding="UTF-8" ?>
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://www.sitemaps.org/schemas/sitemap/0.9 http://www.sitemaps.org/schemas/sitemap/0.9/sitemap.xsd">
  <!-- created with Free Online Sitemap Generator www.xml-sitemaps.com -->
  <url>
    <loc>https://vit.ac.in/</loc>
    <lastmod>2024-11-05T05:35:29+00:00</lastmod>
    <priority>1.00</priority>
  </url>
  <url>
    <loc>https://vit.ac.in/events-photos-and-videos/</loc>
    <lastmod>2024-11-05T05:35:29+00:00</lastmod>
    <priority>0.80</priority>
  </url>
  <url>
    <loc>https://vit.ac.in/admissions/international/overview/</loc>
    <lastmod>2024-11-05T05:35:29+00:00</lastmod>
    <priority>0.80</priority>
  </url>
  <url>
    <loc>https://vit.ac.in/about-vit/</loc>
    <lastmod>2024-11-05T05:35:29+00:00</lastmod>
    <priority>0.80</priority>
  </url>
  <url>
    <loc>https://vit.ac.in/about-vision-mission/</loc>
    <lastmod>2024-11-05T05:35:29+00:00</lastmod>
    <priority>0.80</priority>
  </url>
  <url>
    <loc>https://vit.ac.in/vit-milestones/</loc>
    <lastmod>2024-11-05T05:35:29+00:00</lastmod>
    <priority>0.80</priority>
  </url>
  <url>
    <loc>https://vit.ac.in/about/leadership/</loc>
    <lastmod>2024-11-05T05:35:29+00:00</lastmod>
    <priority>0.80</priority>
  </url>
  <url>
    <loc>https://vit.ac.in/governance/</loc>
    <lastmod>2024-11-05T05:35:29+00:00</lastmod>
    <priority>0.80</priority>
  </url>
  <url>
    <loc>https://vit.ac.in/about/administrative-offices/</loc>
    <lastmod>2024-11-05T05:35:29+00:00</lastmod>
    <priority>0.80</priority>
  </url>
  <url>
    <loc>https://vit.ac.in/about/infrastructure/</loc>
    <lastmod>2024-11-05T05:35:29+00:00</lastmod>
    <priority>0.80</priority>
  </url>
  <url>
    <loc>https://vit.ac.in/about/sustainability/</loc>
    <lastmod>2024-11-05T05:35:29+00:00</lastmod>
    <priority>0.80</priority>
  </url>
  <url>
    <loc>https://vit.ac.in/true-green/</loc>
    <lastmod>2024-11-05T05:35:29+00:00</lastmod>
  </url>
</urlset>
```

Screenshot 7: sitemap.xml — lists all indexed pages with last-modified dates

Part G: Overall Observations & Conclusion

Key Findings Summary

Area	Finding	Risk Level
Hosting	Tata Communications — single IP, no redundancy	Medium
DNS	Amazon Route 53 — well managed, redundant	Low
CMS	WordPress — confirmed via robots.txt	Medium-High
Email	Google Workspace + Microsoft 365 + Amazon SES + Cisco ESA	Low
Security Headers	None of the major headers detected	Medium-High
HTTPS	Active — GlobalSign SSL certificate	Low (Good)
robots.txt	Present — reveals /wp-admin/ and /xmlrpc.php	Medium-High
Sitemap	Present — full page index exposed	Low
DNSSEC	Not enabled	Medium

Conclusion

The passive reconnaissance on VIT University's website (www.vit.ac.in) revealed a comprehensive picture of the institution's web infrastructure using only publicly available tools and data.

The most significant finding was the confirmation of WordPress CMS through the robots.txt file, which explicitly lists /wp-admin/ and /xmlrpc.php. The xmlrpc.php file is a well-documented WordPress attack vector — while blocked from search engine indexing, it may still be accessible and exploitable. This is a common vulnerability in WordPress-based university websites.

The email infrastructure is unusually complex, using four separate providers: Google Workspace (primary), Microsoft 365 (backup), Amazon SES (transactional), and Cisco ESA (security gateway). This level of redundancy is uncommon and suggests VIT takes email reliability seriously.

The absence of all major HTTP security headers (CSP, HSTS, X-Frame-Options, etc.) is a concern, particularly for a WordPress site which is already a high-value target for automated attacks. DNSSEC is also not enabled on the domain.

This exercise demonstrated how much sensitive infrastructure information can be gathered passively — CMS type, hosting provider, email setup, and sensitive paths — all without touching the target system.

Disclaimer: This reconnaissance was performed strictly passively using public tools and open-source data. No systems were accessed, no exploits were attempted, and no sensitive data was retrieved. This report is submitted as part of the White Band Associates Ethical Hacking Internship Task 01.